BSs Computer Science
Probability and statistics - 1st semester - Mid term

November 10, 2025

Please, detail your reasoning as much as possible. Total of points 22.

Use of phones, documents, or any form of cheating is strictly prohibited. Any student caught cheating will receive a grade of 0.

Exercise 1: lecture questions (4 pts)

0.5 pts per question

1. What happens to the mean and median if one extreme value (outlier) is added to the dataset?
2. When you want to compare values between different categories (for example, average income by region or sales by product type), which type of diagram is the most appropriate to use, and why?
3. Explain the difference between a percentile and a percentage.
4. For ordinal data (e.g., satisfaction from 1 = “Very dissatisfied” to 5 = “Very satisfied”), which measure(s) of central tendency are appropriate, and why?
5. What is the main difference between variance and standard deviation?
6. Give at least one true statement if A and B are independents.
7. In a symmetric distribution (like the normal distribution), what is the relationship between mean and median?
8. What the main difference between Permutation and Combination?

Exercise 2: Descriptive Statistics and Outlier Detection (3 pts)

A data analyst records the number of messages received each hour during one workday (10 hours):

12, 15, 9, 8, 20, 40, 13, 11, 5, 35

1. Compute the **sample mean** $\overline{x}$ and median. What do you observe about this distribution ? (0.5 pt)
2. Compute the **sample variance** and **sample standard deviation** (0.5 pt)
3. Determine the **five-number summary** (0.75 pts)
4. Compute the **IQR**. Are they any outliers ? Detail your reasoning. (1 pt)
5. Draw a box plot freehand and place all the important elements on the graph (0.25 pt)

Exercise 3: Probability Rules and Inclusion–Exclusion (4 pts)

A survey was conducted among 200 university students about their beverage preferences. It was found that 120 of them said they like tea (**T**), while 90 reported that they like coffee **C**. Interestingly, 50 students mentioned that they enjoy both tea and coffee. Using this information, answer the following questions.

1. Compute $P(T \cup C)$. Detail your reasoning. (1 pt)
2. Compute the probability that a student likes **tea but not coffee**. (0.5 pt)
3. Compute the probability that a student likes **neither** tea nor coffee. (0.5 pt)
4. Are T and C **independent**? Justify numerically. (1 pt)

Exercise 4: Conditional Probability (2 pts)

A diagnostic test is used to detect a rare disease. About 1% of the population has the disease. If a person has the disease, the test correctly gives a positive result 95% of the time. If a person does not have the disease, the test correctly gives a negative result 99% of the time.

Use this information to answer the following questions.

1. Compute the probability that a randomly selected person will test positive. (1 pt)
2. Out of all people who test positive, what proportion actually have the disease? Explain your reasoning. (1 pt)

Exercise 5: Z-scores and Normal Distribution (3 pts)

Men's heights are normally distributed with a mean of 175 cm and a standard deviation of 7 cm, while women's heights are normally distributed with a mean of 162 cm and a standard deviation of 6 cm. That is,

$$X_m \sim N(175, 7^2) \quad \text{and} \quad X_w \sim N(162, 6^2).$$

1. Compute the **Z-score (for men)** for x = 180 cm. What does a Z-score tell us about a value?. (1 pt)
2. Using a standard normal table, compute the percentage of men that don't need to duck when walking through a door that is 180 cm high? (1 pt)
3. What percentage of women must duck when walking through a door that is 180 cm high? (1 pt)

Standard Normal (Z) Table:

The table below gives $P(Z < z)$ for a standard normal variable.

z	0	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
+0	.50000	.50399	.50798	.51197	.51595	.51994	.52392	.52790	.53188	.53586
+0.1	.53983	.54380	.54776	.55172	.55567	.55966	.56360	.56749	.57142	.57535
+0.2	.57926	.58317	.58706	.59095	.59483	.59871	.60257	.60642	.61026	.61409
+0.3	.61791	.62172	.62552	.62930	.63307	.63683	.64058	.64431	.64803	.65173
+0.4	.65542	.65910	.66276	.66640	.67003	.67364	.67724	.68082	.68439	.68793
+0.5	.69146	.69497	.69847	.70194	.70540	.70884	.71226	.71566	.71904	.72240
+0.6	.72575	.72907	.73237	.73565	.73891	.74215	.74537	.74857	.75175	.75490
+0.7	.75804	.76115	.76424	.76730	.77035	.77337	.77637	.77935	.78230	.78524
+0.8	.78814	.79103	.79389	.79673	.79955	.80234	.80511	.80785	.81057	.81327
+0.9	.81594	.81859	.82121	.82381	.82639	.82894	.83147	.83398	.83646	.83891
+1	.84134	.84375	.84614	.84849	.85083	.85314	.85543	.85769	.85993	.86214
+1.1	.86433	.86650	.86864	.87076	.87286	.87493	.87698	.87900	.88100	.88298
+1.2	.88493	.88686	.88877	.89065	.89251	.89435	.89617	.89796	.89973	.90147
+1.3	.90320	.90490	.90658	.90824	.90988	.91149	.91308	.91466	.91621	.91774
+1.4	.91924	.92073	.92220	.92364	.92507	.92647	.92785	.92922	.93056	.93189
+1.5	.93319	.93448	.93574	.93699	.93822	.93943	.94062	.94179	.94295	.94408
+1.6	.94520	.94630	.94738	.94845	.94950	.95053	.95154	.95254	.95352	.95449
+1.7	.95543	.95637	.95728	.95818	.95907	.95994	.96080	.96164	.96246	.96327
+1.8	.96407	.96485	.96562	.96638	.96712	.96784	.96856	.96926	.96995	.97062
+1.9	.97128	.97193	.97257	.97320	.97381	.97441	.97500	.97558	.97615	.97670
+2	.97725	.97778	.97831	.97882	.97932	.97982	.98030	.98077	.98124	.98169
+2.1	.98214	.98257	.98300	.98341	.98382	.98422	.98461	.98500	.98537	.98574
+2.2	.98610	.98645	.98679	.98713	.98745	.98778	.98809	.98840	.98870	.98899
+2.3	.98928	.98956	.98983	.99010	.99036	.99061	.99086	.99111	.99134	.99158
+2.4	.99180	.99202	.99224	.99245	.99266	.99286	.99305	.99324	.99343	.99361
+2.5	.99379	.99396	.99413	.99430	.99446	.99461	.99477	.99492	.99506	.99520
+2.6	.99534	.99547	.99560	.99573	.99585	.99598	.99609	.99621	.99632	.99643
+2.7	.99653	.99664	.99674	.99683	.99693	.99702	.99711	.99720	.99728	.99736
+2.8	.99744	.99752	.99760	.99767	.99774	.99781	.99788	.99795	.99801	.99807
+2.9	.99813	.99819	.99825	.99831	.99836	.99841	.99846	.99851	.99856	.99861
+3	.99865	.99869	.99874	.99878	.99882	.99886	.99889	.99893	.99896	.99900
+3.1	.99903	.99906	.99910	.99913	.99916	.99918	.99921	.99924	.99926	.99929
+3.2	.99931	.99934	.99936	.99938	.99940	.99942	.99944	.99946	.99948	.99950
+3.3	.99952	.99953	.99955	.99957	.99958	.99960	.99961	.99962	.99964	.99965
+3.4	.99966	.99968	.99969	.99970	.99971	.99972	.99973	.99974	.99975	.99976
+3.5	.99977	.99978	.99978	.99979	.99980	.99981	.99981	.99982	.99983	.99983
+3.6	.99984	.99985	.99985	.99986	.99986	.99987	.99987	.99988	.99988	.99989
+3.7	.99989	.99990	.99990	.99990	.99991	.99991	.99992	.99992	.99992	.99992
+3.8	.99993	.99993	.99993	.99994	.99994	.99994	.99994	.99995	.99995	.99995
+3.9	.99995	.99995	.99996	.99996	.99996	.99996	.99996	.99996	.99997	.99997
+4	.99997	.99997	.99997	.99997	.99997	.99997	.99998	.99998	.99998	.99998

Exercise 6: Lottery and Probability (2 pts)

In a national lottery, 6 numbers are drawn at random from 1 to 49, without replacement. A player chooses 6 numbers in advance.

1. How many different combinations of 6 numbers can be drawn? (0.5 pt)
2. What is the probability that a player's 6 numbers match the 6 winning numbers exactly? (0.5 pt)
3. What is the probability that the player matches exactly 3 winning numbers? (1 pt)

Exercise 7: Permutations with Repeated Letters and Probability (2 pts)

Consider the word **ANTICONSTITUTIONNELLEMENT**, which contains 25 letters in total. For both questions, write only the explicit formula, do not give numerical value.

1. Write an expression for the number of distinct "words" that can be formed by rearranging all the letters. (1 pt)
2. In one random arrangement, what is the probability that all the **T**'s appear next to each other (as a single block) (1 pt)

Exercise 8: Critical Thinking in Statistics (3 pts)

Read the following study and examine the figures carefully. Identify at least two major biases or methodological and statistical errors, and for each:

1- Explain why it is problematic. (1.5 pt)

2- Propose how the study or visualization could be improved. (1.5 pt)

Does Listening to Music While Studying Improve Exam Performance?

Study description

In recent years, social media has been full of videos and articles claiming that listening to background music improves concentration and academic performance. Curious about this trend, a small group of independent student researchers decided to test whether studying with music truly leads to better grades.

They designed an online survey distributed through TikTok, and WhatsApp student communities across Île-de-France. In total, 212 undergraduate students aged between 18 and 25 participated voluntarily.

Each participant answered a short questionnaire asking:

- Whether they usually study with music or in silence
- Their average exam grade from the previous semester (on a 0–20 scale).

Based on their answers, students were divided into two groups:

- Music while studying (n = 138)
- Silence while studying (n = 74)

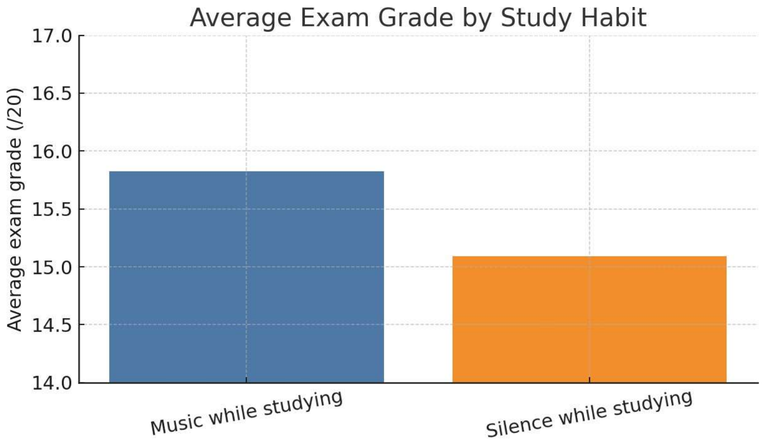
The researchers compared the mean exam grade between the two groups. They did not report any measure of variability and did not specify whether grades followed a normal distribution.

Results

The average self-reported exam grades were:

- Music group: 16.2 / 20
- Silence group: 14.8 / 20

The results were visualized in the following figures provided in the study.



Conclusion (as written by the authors)

“Our findings clearly show that students who listen to music while studying achieve better academic results. This demonstrates that listening to music improves exam performance.